Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.3584659108907591, median 0.26127271692304943, std: 0.31872865710520715
Gyroscope error (imu0): mean 0.2929178700360966, median 0.16045558211779193, std: 0.4881330432606213
Accelerometer error (imu0): mean 0.33268104145119787, median 0.18969133967949073, std: 0.6089863059232333

Residuals

Reprojection error (cam0) [px]: mean 0.3584659108907591, median 0.26127271692304943, std: 0.31872865710520715
Gyroscope error (imu0) [rad/s]: mean 0.011565611930998303, median 0.0063354516223544976, std: 0.01927351632167011
Accelerometer error (imu0) [m/s^2]: mean 0.12916272409433294, median 0.07364726905756244, std: 0.23643766974538669

Transformation (cam0):

T_ci: (imu0 to cam0):
[[-0.22712634 -0.94936432 0.21707375 -0.0347539]
 [0.40492843 -0.29477959 -0.86552756 0.01730295]
 [0.8856899 -0.10868477 0.45137681 -0.01429993]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[-0.22712634 0.40492843 0.8856899 -0.00223467]
 [-0.94936432 -0.29477959 -0.10868477 -0.02944775]
 [0.21707375 -0.86552756 0.45137681 0.02897499]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.008456274899756705

Gravity vector in target coords: [m/s^2]
[-0.15874248 -9.35609659 2.93371443]

Calibration configuration
==========

cam0

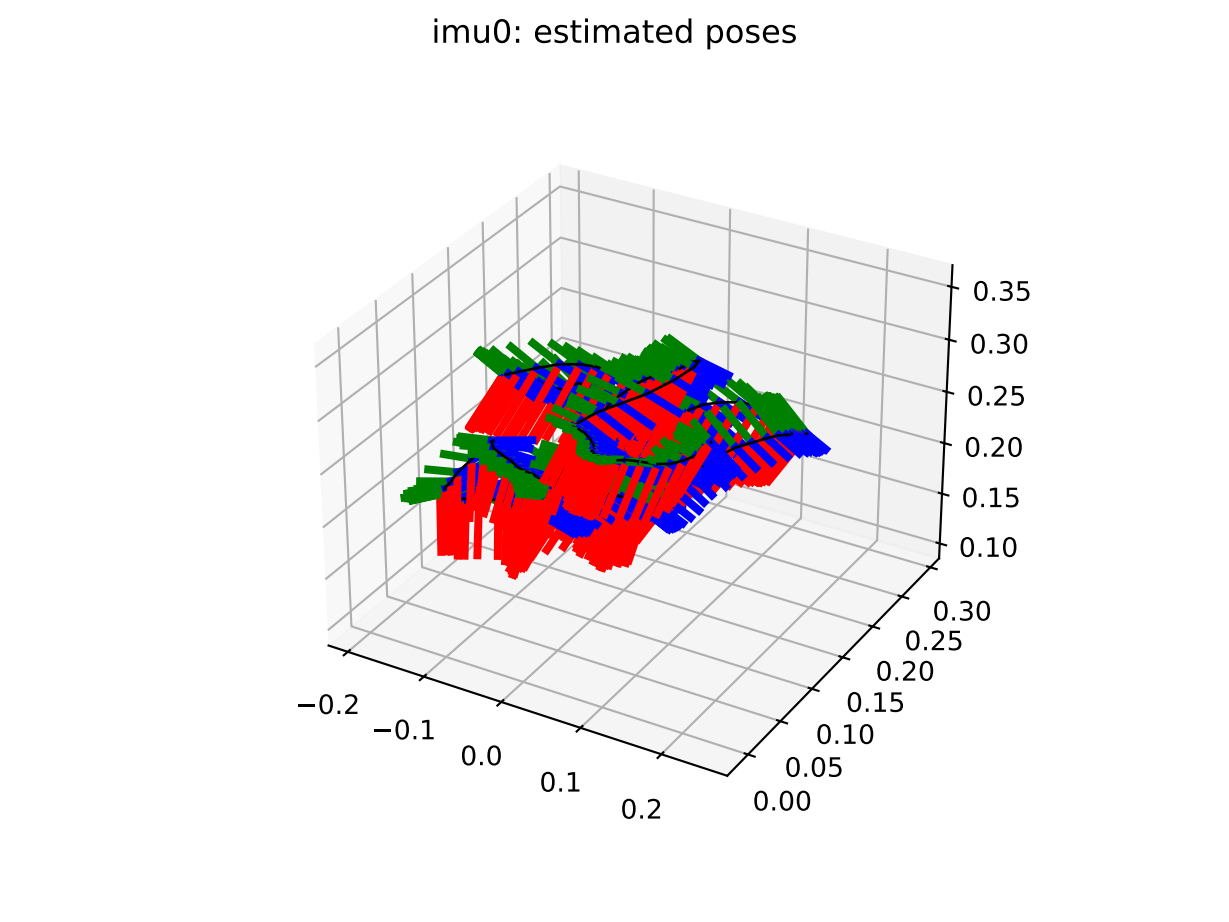
  Camera model: pinhole
  Focal length: [652.879962242561, 652.6913800309958]
  Principal point: [402.7459765146756, 293.51689967360625]
  Distortion model: radtan
  Distortion coefficients: [0.19004630188120233, -0.3566296810567926, -0.001643546078163314,
0.0007208195618292788]
  Type: aprilgrid
  Tags:
    Rows: 6
    Cols: 6
    Size: 0.017733333333333334 [m]
    Spacing 0.00532 [m]

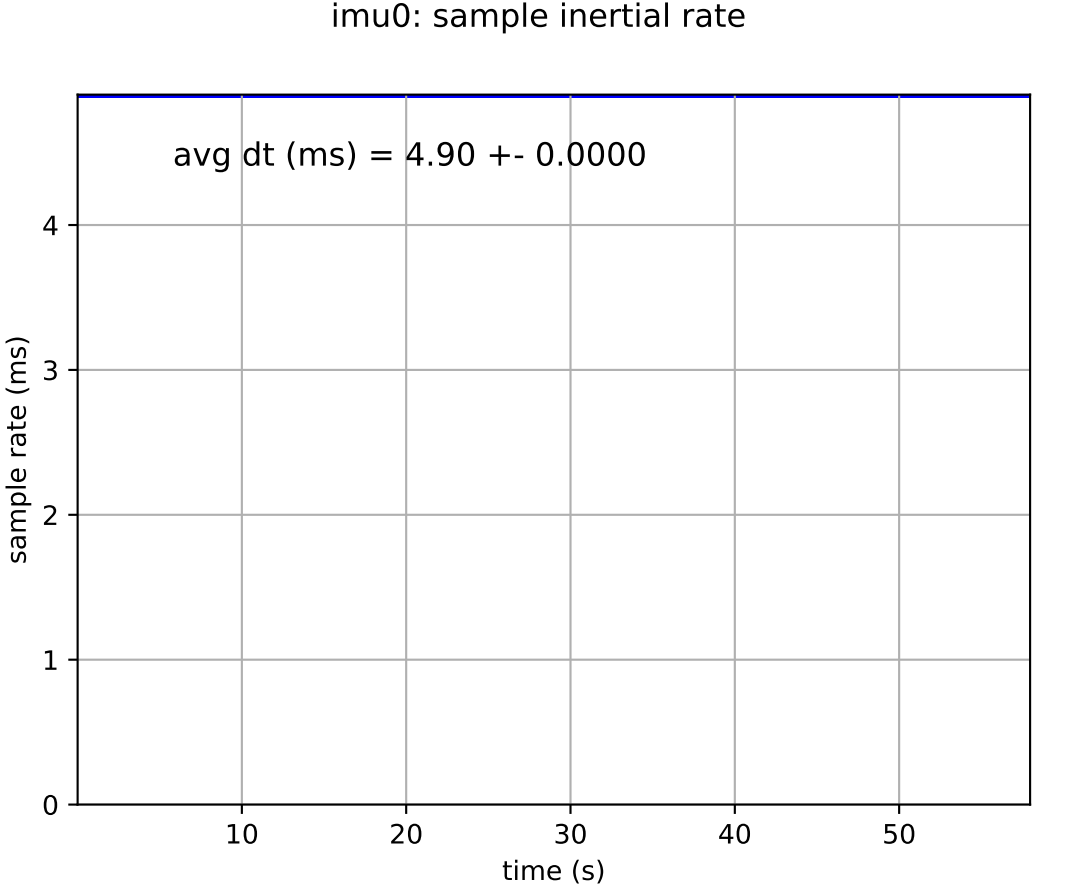
IMU configuration
==========

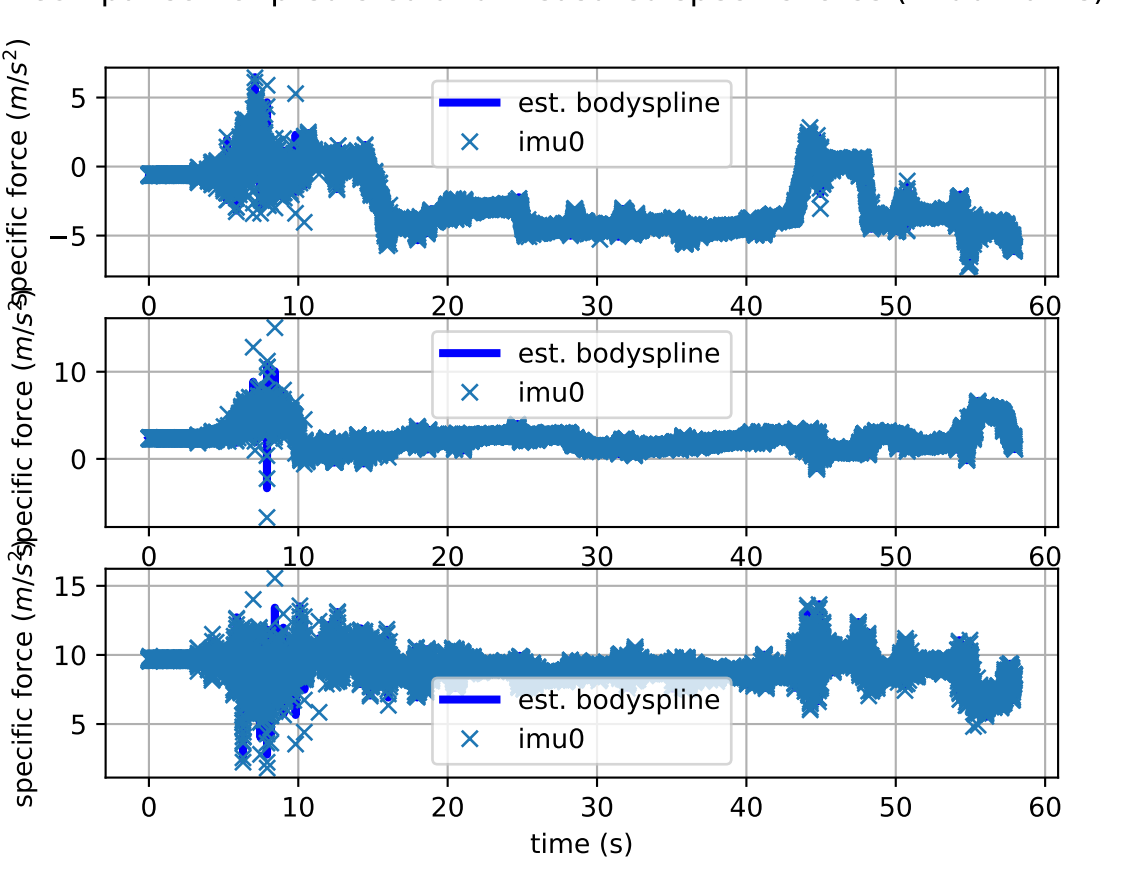
IMU0:

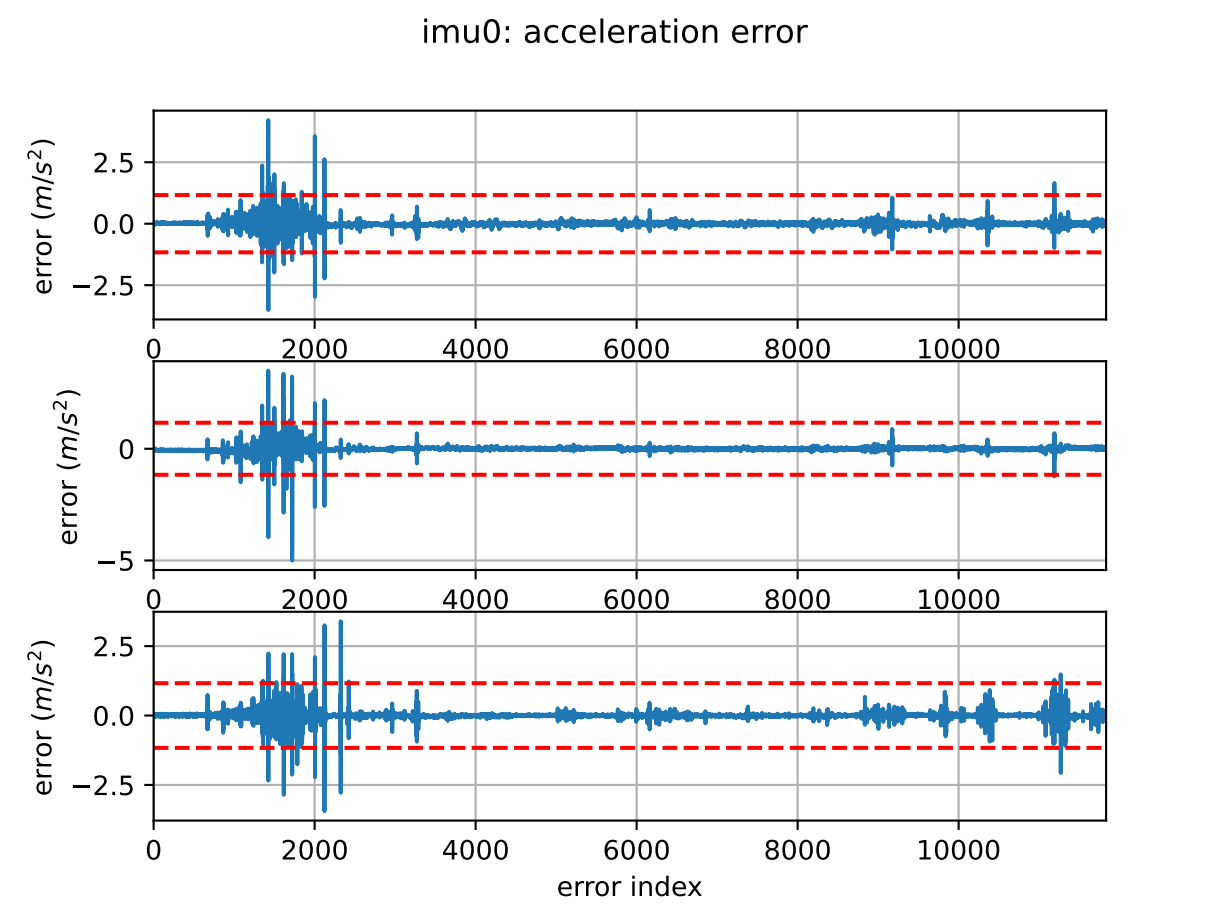
  Model: calibrated
  Update rate: 204.090470950253
  Accelerometer:
    Noise density: 0.027176766288825538
    Noise density (discrete): 0.38824792519257595
    Random walk: 0.0067023621585376265
  Gyroscope:
    Noise density: 0.002763830369432253
    Noise density (discrete): 0.039484145947029654
    Random walk: 0.00014770260860981916

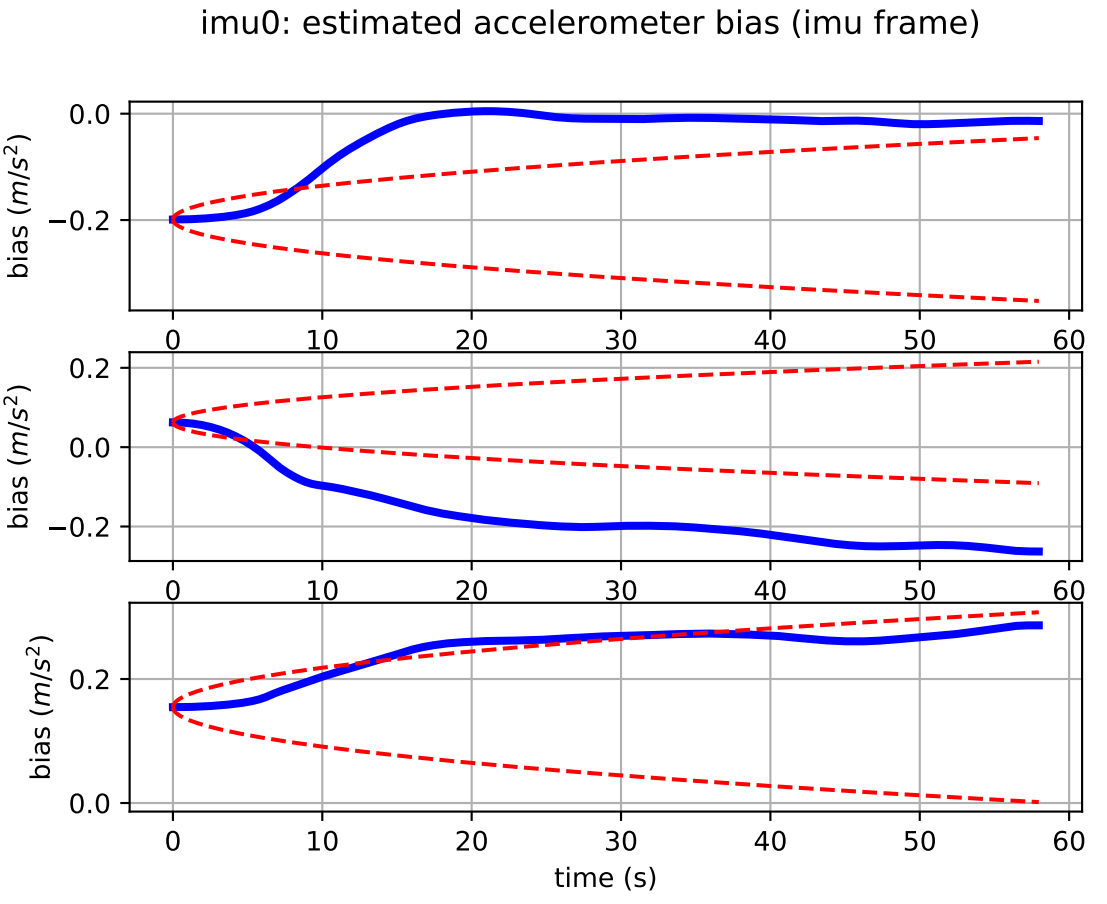
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

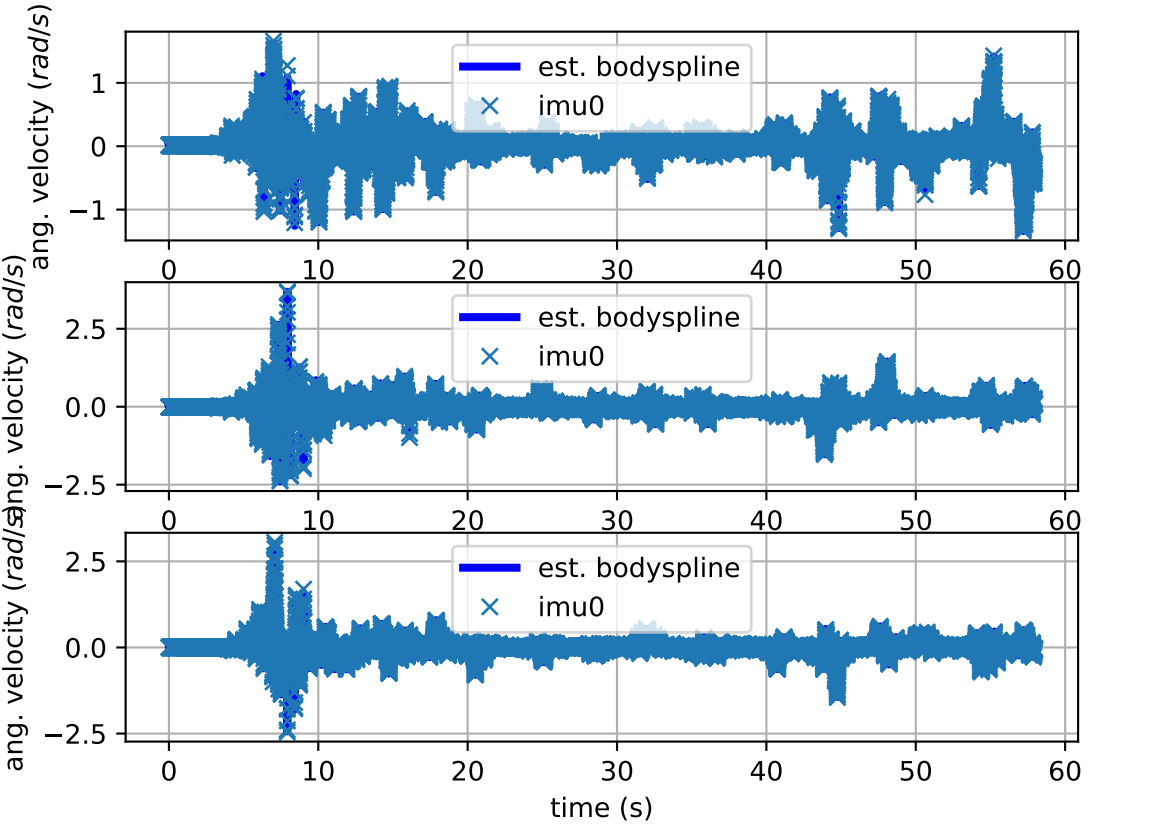

imu0: estimated poses
0.35
0.30
0.25
0.20
0.15
0.10
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.2
−0.1
0.0
0.1
0.2

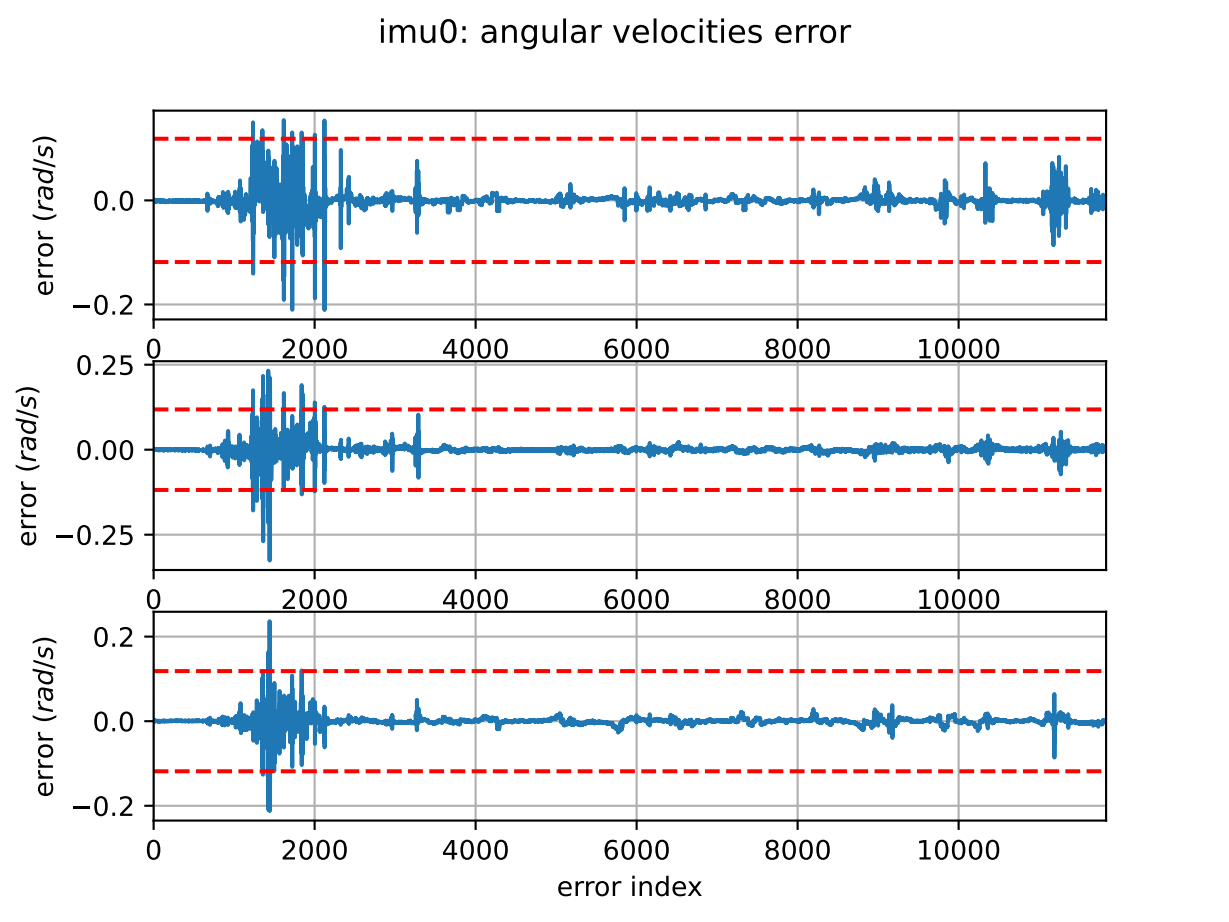

imu0: sample inertial rate
avg dt (ms) = 4.90 +- 0.0000
sample rate (ms)
0
1
2
3
4
10
20
30
40
50
time (s)

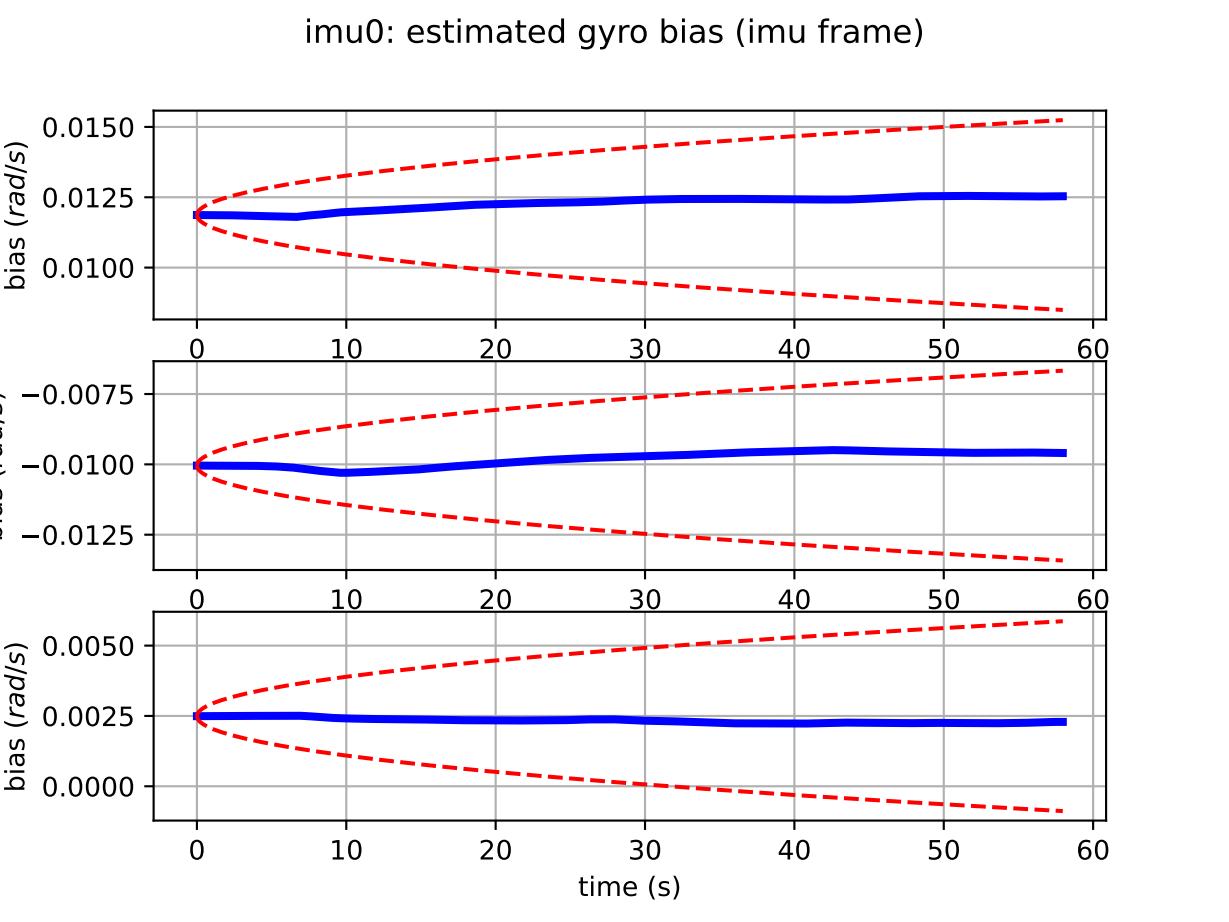

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
specific force (m/s^2)
specific force (m/s^2)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
time (s)

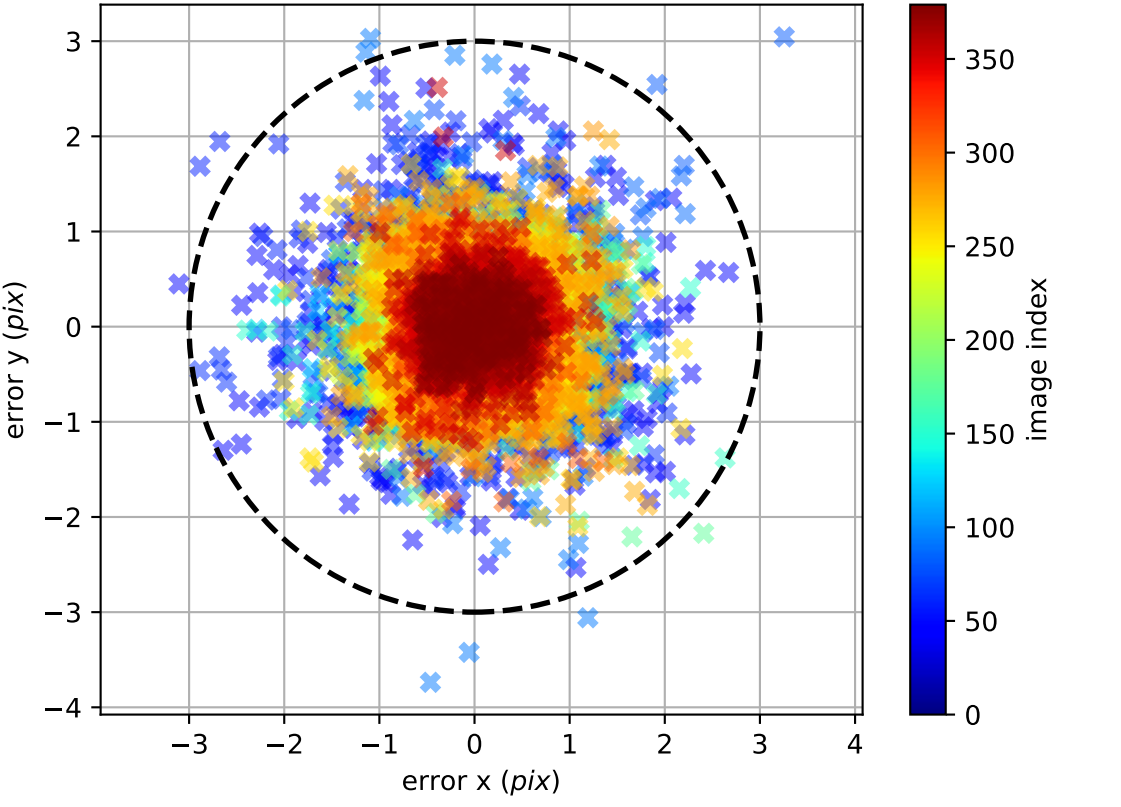

imu0: acceleration error
error (m/s^2)
2.5
0.0
−2.5
0
2000
4000
6000
8000
10000
error (m/s^2)
0
−5
0
2000
4000
6000
8000
10000
error (m/s^2)
2.5
0.0
−2.5
0
2000
4000
6000
8000
10000
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.0
−0.2
0
10
20
30
40
50
60
bias (m/s^2)
0.2
0.0
−0.2
bias (m/s^2)
0.2
0.0
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
time (s)


imu0: angular velocities error
error (rad/s)
0.0
−0.2
0
2000
4000
6000
8000
10000
error (rad/s)
0.25
0.00
−0.25
0
2000
4000
6000
8000
10000
error (rad/s)
0.2
0.0
−0.2
0
2000
4000
6000
8000
10000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.0150
0.0125
0.0100
−0.0075
−0.0100
−0.0125
0.0050
0.0025
0.0000
0
10
20
30
40
50
60
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
3
2
1
0
−1
−2
−3
−4
−3
−2
−1
0
1
2
3
4
350
300
250
200
150
100
50
0